



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Denkleiers • Leading Minds • Dikgopolo tša Dihlalefi

Department of Computer Science
University of Pretoria
Digital Forensics and Investigations
COS783
Practical 1
Student number: u25757637
E-mail address: u25757637@tuks.ac.za

25757637

20 August 2025

Introduction

This investigation examined a forensic image from a university-issued Windows 10 device suspected of policy violations. The objective was to identify traces of unauthorized file sharing, malware execution, and anonymized browsing.

Scope

The process involved verifying the image's integrity, extracting timeline and browser artifacts, identifying suspicious software, and analyzing persistence mechanisms and deleted data. All procedures were conducted using approved forensic tools within a secure, controlled environment.

Methodology

The forensic process began with validating the integrity of the acquired image using SHA-256 hashing. The image was then mounted in a controlled analysis environment configured with Autopsy and PowerShell. Timeline analysis was conducted to reconstruct user activity, focusing on browser history, file access patterns, and system events. Artifacts related to anonymized browsing, torrent usage, and executable files were extracted and examined. Registry entries, startup configurations, and deleted file traces were analyzed to identify persistence mechanisms and concealment strategies. All findings were documented with timestamps and verified against system logs to ensure accuracy and reproducibility.

Task 1: Preparation and Verification of Image Integrity

1.1. Hashes of the forensic image provided using both md5sum and sha256sum

md5sum command

```
Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm MD5
```

sha256 command

```
Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256
```

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd "D:\COSS 783\MSEdge - COS783-disk1.vdi"
PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         92DC6A56E2BAB4F6C2426A9E59AC796D8F09DE5FADA61B4EFD132777FB8D2744
Path
-----
D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi

PS D:\COSS 783\MSEdge - COS783-disk1.vdi>
PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm MD5

Algorithm      Hash
-----
MD5            649872C3C3AE98E787FA2211338793A8
Path
-----
D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi

PS D:\COSS 783\MSEdge - COS783-disk1.vdi> Get-FileHash "MSEdge - COS783-disk1.vdi" -Algorithm SHA256

Algorithm      Hash
-----
SHA256         92DC6A56E2BAB4F6C2426A9E59AC796D8F09DE5FADA61B4EFD132777FB8D2744
Path
-----
D:\COSS 783\MSEdge - COS783-disk1.vdi\MSEdge - COS783-disk1.vdi
```

Figure 1: md5sum and sha256sum output on Powershell

1.2. Importance of hashing in digital forensics

1. Evidence Integrity

Hashing creates a unique digital signature to confirm that forensic data remains unchanged during acquisition, analysis, or transfer.

2. Chain of Custody

Tracking hash values at each stage ensures a verifiable custody trail, proving the evidence is identical to its original form for legal use.

1.3. Image Integrity

The forensic image was verified using the `sha256sum` utility in PowerShell. The resulting hash value was:

92dc6a56e2bab4f6c2426a9e59ac796d8f09de5fada61b4efd132777fb8d2744

Hashing is essential in digital forensics as it ensures the integrity of evidence by producing a unique digital fingerprint. Any alteration to the file would result in a different hash, thereby alerting investigators to potential tampering. The image's integrity was confirmed, and no discrepancies were found.

Task 2: Forensic Acquisition

2.1. Loading the image on Autopsy(the chosen tool)

The `MSEdge - COS783-disk1.vdi` was converted to the raw format, and then the raw image was ingested instead. Autopsy handles raw images much more

consistently.

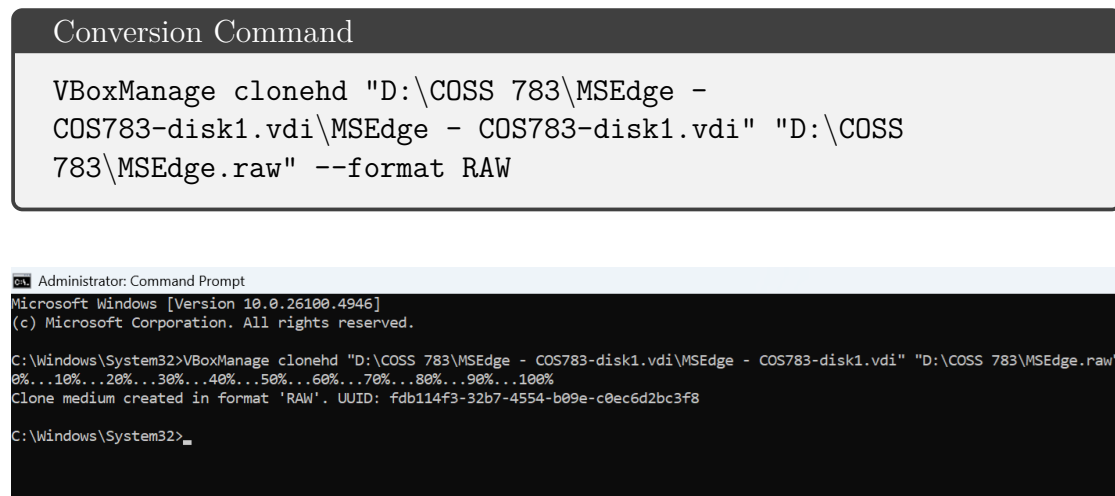


Figure 2: Successful .raw file saved to COSS 783 folder

SHA256 hashes were computed for both the .raw and .vdi disk images. The differing hash values on the following console window confirm that the files are structurally distinct, validating the integrity of each format for its respective forensic purpose.

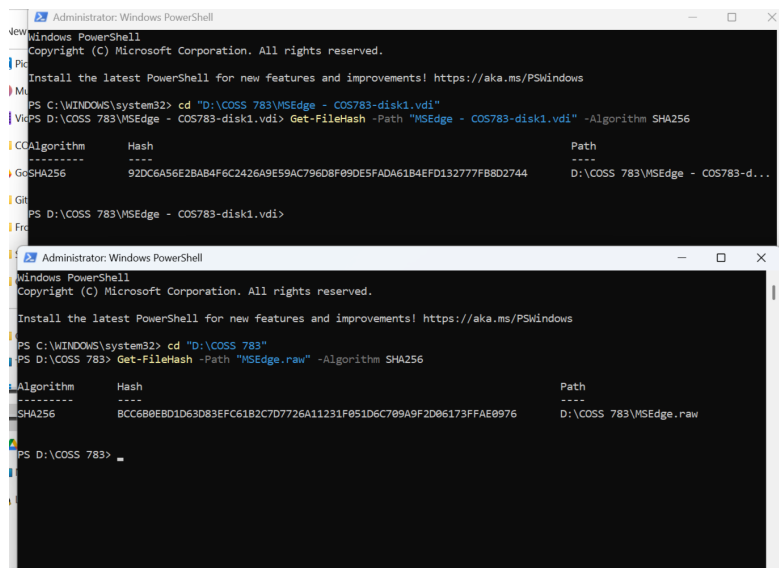
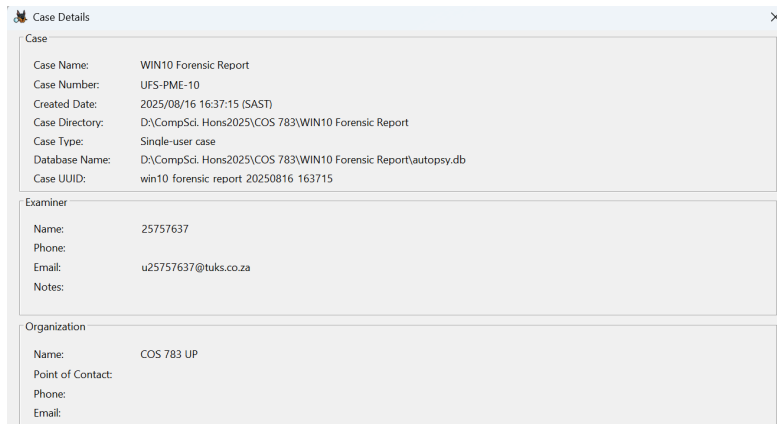


Figure 3: .raw and .vdi disk image hashes

Data source

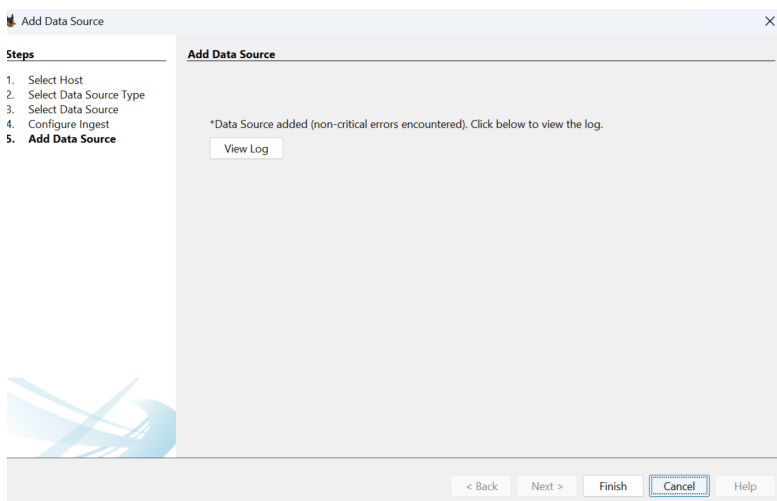


Case	
Case Name:	WIN10 Forensic Report
Case Number:	UFS-PME-10
Created Date:	2025/08/16 16:37:15 (SAST)
Case Directory:	D:\CompSci. Hons2025\COS 783\WIN10 Forensic Report
Case Type:	Single-user case
Database Name:	D:\CompSci. Hons2025\COS 783\WIN10 Forensic Report\autopsy.db
Case UUID:	win10 forensic report 20250816 163715

Examiner	
Name:	25757637
Phone:	
Email:	u25757637@tuks.co.za
Notes:	

Organization	
Name:	COS 783 UP
Point of Contact:	
Phone:	
Email:	

Figure 4: Digital Forensics Case Details



Steps

1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. **Add Data Source**

Add Data Source

*Data Source added (non-critical errors encountered). Click below to view the log.

[View Log](#)

< Back Next > Finish **Cancel** Help

Figure 5: Successful data source loading

After creating a new case on Autopsy and adding the data source, MSEdge.raw, the following ingest modules were enabled:

- Recent Activity
- Operating System Information
- File Type Identification

- Extension Mismatch Detector
- Embedded File Extractor
- Keyword Search
- Email Parser
- Encryption Detection
- Interesting Files Identifier
- Virtual Machine Extractor
- Data Source Integrity

2.2. System Metadata

a) Hostname: MSEDGEWIN10

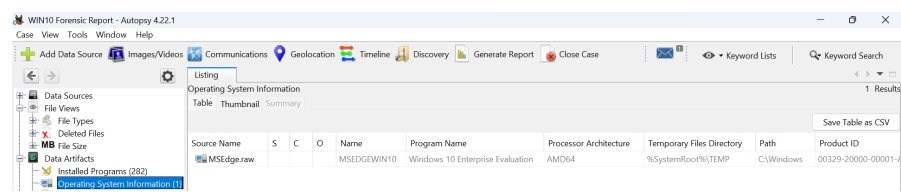


Figure 6: Data Artifact = OS Information

b) User Accounts

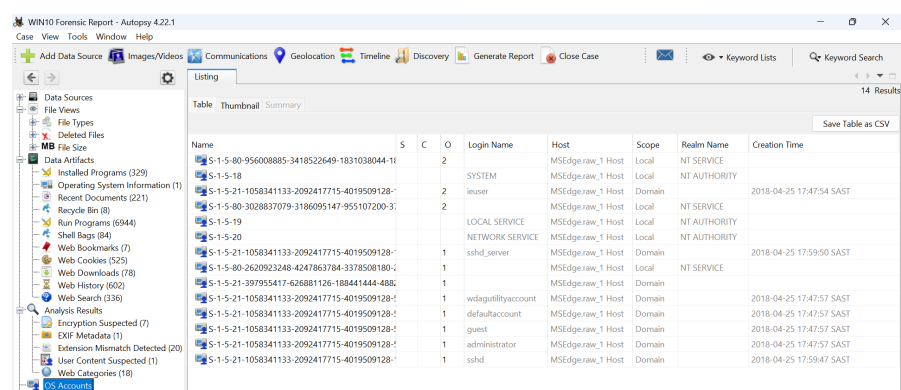
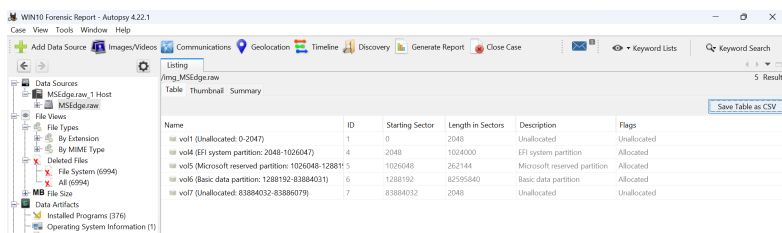


Figure 7: OS Accounts

c) Last Shutdown Time

2.3 File system and Partition structure



Name	ID	Starting Sector	Length in Sectors	Description	Flags
vol1 (Unallocated: 0-2047)	1	0	2048	Unallocated	Unallocated
vol4 (EFI system partition: 2048-1026047)	4	2048	1024000	EFI system partition	Allocated
vol5 (Microsoft reserved partition: 1026048-1288119)	5	1026048	262144	Microsoft reserved partition	Allocated
vol6 (Basic data partition: 1288192-83884031)	6	1288192	82595840	Basic data partition	Allocated
vol7 (Unallocated: 83884032-83886079)	7	83884032	2048	Unallocated	Unallocated

Figure 8: Output obtained from the disk image

Partitioning Scheme: GPT (based on presence of EFI and Microsoft reserved partitions)

Total Partitions: 5

- vol1: Unallocated (sectors 0–2047)
- vol4: EFI System Partition (sectors 2048–1026047)
- vol5: Microsoft Reserved Partition (sectors 1026048–1288191)
- vol6: Basic Data Partition (sectors 1288192–83884031)
- vol7: Unallocated (sectors 83884032–83886079)

Primary File System: NTFS (used in vol6, the main Windows partition)

Interpretation: The disk image uses a GPT layout with standard Windows partitions. The NTFS file system is used for the main data volume, typical of Windows 10 installations.

Task 3: Timeline Analysis and User Activity

3.1. When the suspicious activity started

The following are the filters used on the timeline to search for suspicious activity:

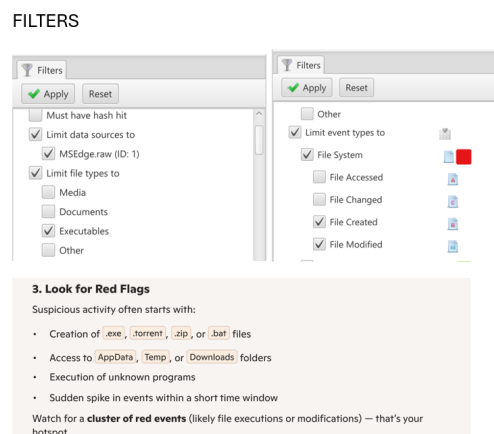


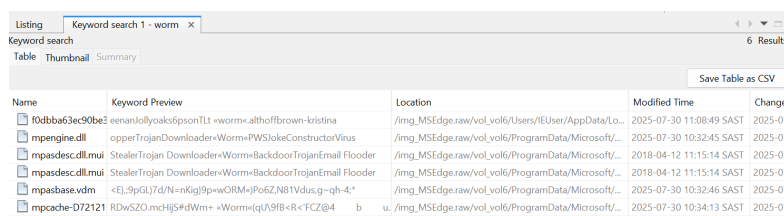
Figure 9: Filters

Suspicious Activity Timeline

Date	Event	Evidence	Interpretation
2018-01-25	Execution of <code>sdelete.exe</code>	Found in 2018 <code>Jan.csv</code>	First sign of anti-forensics; secure deletion tool often used to erase traces
2018-02-13	VirtualBox Guest Additions installed	Found in 2018 <code>Feb.csv</code>	Suggests VM setup — could be attacker sandbox or analyst environment
2018-08-30	Malware indicators in DLLs and cache files	<code>mpengine.dll</code> , <code>mpasdesc.dll.mui</code> , <code>f0dbba63ec90be3</code>	TrojanDownloader, Worm, Backdoor behavior; possible data staging or exfiltration
2018 (undated)	Legacy registry artifact	<code>ntuser.dat.LOG1</code>	May reveal historical user behavior or software installs

2025-07-30 11:08:49	Chrome extension with “eraser” keyword	Located in <code>component_crx_cache</code>	Suggests privacy-focused or wiping extension; identical timestamps — possible anti-forensic intent
2025-07-30 12:16:40	Deletion of <code>utorrent_- installer.exe</code>	Recycle Bin artifact <code>\$R00KXBM.exe</code>	Manual deletion linked to torrent activity; user attempted concealment without secure wipe
2025-07-30	CCleaner keyword in Avast directory	Timestamps zeroed	Indicates prior use or detection of anti-forensic tool; metadata wiping suspected
2025-07-30	Access to Chrome Secure Preferences & Local State	Browser config files	Confirms browser usage; may contain encrypted traces of extensions or privacy tools
2025-07-30	Coordinated DLL and cache file modifications	Found in 2025 <code>July18.csv</code> & <code>July18b.csv</code>	Malware reactivation or persistence; mirrors 2018 indicators

”worm” key search



The screenshot shows a web-based keyword search interface. At the top, there's a search bar with the text 'Keyword search 1 - worm'. Below the search bar, there are tabs for 'Table', 'Thumbnail', and 'Summary'. The 'Table' tab is selected, displaying a table with 6 results. The table has columns for Name, Keyword, Preview, Location, Modified Time, and Change. The results are as follows:

Name	Keyword	Preview	Location	Modified Time	Change
f0dbba63ec90be2	eananlojloakspsonTlU	«worm»althoffbrown-kristina	/img_MSEdge.raw/vol_vol6/Users/IEUser/AppData/Lo...	2025-07-30 11:08:49 SAST	2025-07-30 11:08:49 SAST
mpengine.dll	opperTrojanDownloader«Worm»PWSlokeConstructorVirus		/img_MSEdge.raw/vol_vol6/ProgramData/Microsoft/...	2025-07-30 10:32:45 SAST	2025-07-30 10:32:45 SAST
mpasdesc.dll.mui	StealerTrojan Downloader«Worm»BackdoorTrojanEmail Flooder		/img_MSEdge.raw/vol_vol6/ProgramData/Microsoft/...	2018-04-12 11:15:14 SAST	2025-07-30 10:32:45 SAST
mpasbase.vdm	«EJ_9pGj7d/N=nKig9p-wORM»Po6Z.N81Vdus.g-qh-4*»		/img_MSEdge.raw/vol_vol6/ProgramData/Microsoft/...	2018-04-12 11:15:14 SAST	2025-07-30 10:32:45 SAST
mpcache-D72121	RDwSZO.mchjS#dWm+ «Worm»(qU9fB«R<FCZ@4 b u,		/img_MSEdge.raw/vol_vol6/ProgramData/Microsoft/...	2025-07-30 10:32:46 SAST	2025-07-30 10:32:46 SAST
				2025-07-30 10:34:13 SAST	2025-07-30 10:34:13 SAST

Figure 10: ”worm” search output

3.2. File downloads (e.g., .torrent or suspicious executables)

- **Torrent Activity:** Multiple .torrent files from YTS.AG and YTS.MX suggest active media downloads.
- **Software Acquisition:** Transmission installer (.msi) and Ubuntu ISOs indicate software-related downloads, possibly for dual-booting or forensic evasion.

- **Zone.Identifier Tags:** All downloads tagged with Zone ID 3, confirming internet origin.
- **Suspicious File:** Obfuscated filename in `/Videos` may represent encrypted payload or container.

Item	Path	Size	File Name
Updating system information (1)	775f4b6c-810f-4b4e-83d7-ad6d4115503d.tmp-zone.Identifier		Users\Elmer\AppData\Local\Temp\775f4b6c-810f-4b4e-83d7-ad6d4115503d.tmp-zone.Identifier
Recent Documents (265)	ebd44fe4-d9e7-4215-95e4-848d1200b208.tmp-zone.Identifier		Users\Elmer\AppData\Local\Temp\ebd44fe4-d9e7-4215-95e4-848d1200b208.tmp-zone.Identifier
Recycle Bin (9)	Capitain America Civil War (2015) [7YS.GS.torrent].zone.Identifier		Users\Elmer\AppData\Local\Roaming\usercontent\Capitain America Civil War (2015) [7YS.GS.torrent].zone.Identifier
Run Programs (3624)	Iron Man (2008).torrent-zone.Identifier		Users\Elmer\AppData\Local\usercontent\Iron Man (2008).torrent-zone.Identifier
Shell Bags (96)	28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier		Users\Elmer\AppData\Local\28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier
Web Bookmarks (6)	28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier		Users\Elmer\Videos\28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier
Web Cookies (102)	28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier		Users\Elmer\Videos\28 Years Later (2020) [7YS.GS.MRNG].torrent-zone.Identifier
Web Downloads (78)	Capitain America Civil War (2015) [7YS.GS.torrent].zone.Identifier		Windows\Capitain America Civil War (2015) [7YS.GS.torrent].zone.Identifier
Web History (188)	Iron Man (2008) [7YS.GS.BuRaf].torrent-zone.Identifier		Windows\Iron Man (2008) [7YS.GS.BuRaf].torrent-zone.Identifier
Web Sitedata (1)	MicrosoftEdgeCookiesBackup.dat-zone.Identifier		Users\Elmer\MicrosoftEdgeCookiesBackup.dat-zone.Identifier
Analysis Results	MicrosoftEdgeSettingsBackup.dat-zone.Identifier		Users\Elmer\MicrosoftEdgeSettingsBackup.dat-zone.Identifier
Encryption Support (2)	Backup.dat-zone.Identifier		Users\Elmer\MicrosoftEdgeSettingsBackup.dat-zone.Identifier
EXP Meddata (1)	Backup.dat-zone.Identifier		Users\Elmer\MicrosoftEdgeSettingsBackup.dat-zone.Identifier
Extension Minidump Detected (2)	ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier	4	Users\Elmer\AppData\Local\ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier
Keyword Hits (58)	ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier	4	Users\Elmer\AppData\Local\ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier
User Content Suspected (1)	ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier	4	Users\Elmer\AppData\Local\ubuntu-20.04-desktop-amd64.iso.torrent-zone.Identifier
Web Categories (26)	transmission-4.0.6-64bit_1.torrent-zone.Identifier	3	Users\Elmer\Downloads\transmission-4.0.6-64bit_1.torrent-zone.Identifier
AC Accounts			
Tags			

Figure 11: Web Downloads Output

The screenshot shows the VirusShare web application interface. At the top, there's a navigation bar with tabs: Hex, Text, Application, Source File Metadata, OS Account, Data Artifacts, Analysis Results, Content, Annotations, and Other Occurrences. The 'Text' tab is active. Below the tabs, there's a search bar and a 'Matches on page' indicator. The main content area displays the text of the selected file, which is a JSON object representing a GitHub release asset. The JSON content is as follows:

```
{
  "url": "https://release-assets.githubusercontent.com/github-production-release-asset/35129809/d9596751-7405-4548-ae80-32ba4c8d8047?r=rs&v=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%3A28&=2025-07-20T11-34A0693a422f8c5d4bdspr=2018-11-09&sv=bdspr=https&size=25.07-301111c3a0693a422f8c5d4attachment%38=1filem3c.Dtransmission-4.0.6-x64.m3c&st=application%2Foctet-stream&moid=9c62d410-5711-43a1-aed4-ab1947a7ab04&bkid=39b6f5a-997b-474b-312b-9515b9bde4eb&=2025-07-20T10%3A51%
```

Figure 12: Analysis of web downloads zoneID

3.3. Any evidence of file deletion or anti-forensic behaviour

- **Recycle Bin Artifact:** `utorrent_installer.exe` manually deleted on 2025-07-30.
- **CCleaner Keyword:** Found in `/ProgramData/Avast Software/` with zeroed timestamps — possible metadata wiping.
- **Eraser Keyword:** Located in Chrome extension cache; identical timestamps suggest single-use install or timestamp manipulation.
- **Browser Artifacts:** Secure Preferences and Local State files accessed on 2025-07-30 — may contain encrypted traces of user activity.

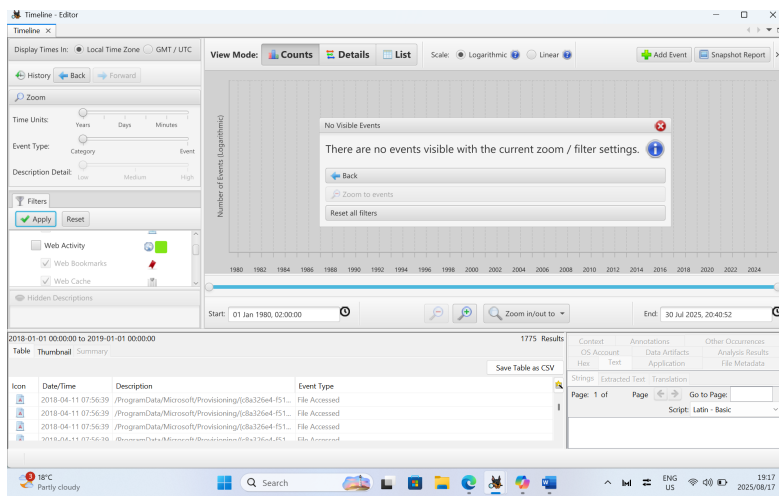


Figure 13: Output of deleted files using filters on the timeline

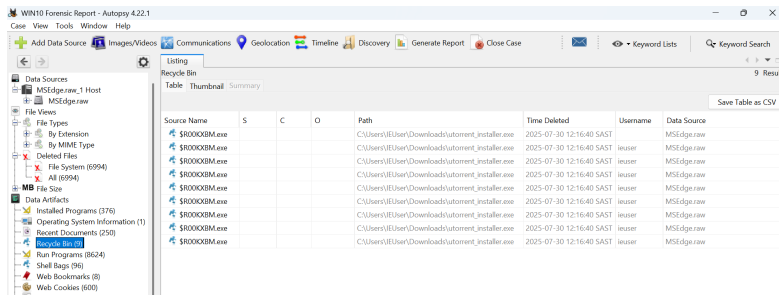


Figure 14: Recycle Bin Output

"CCleaner" Key Search

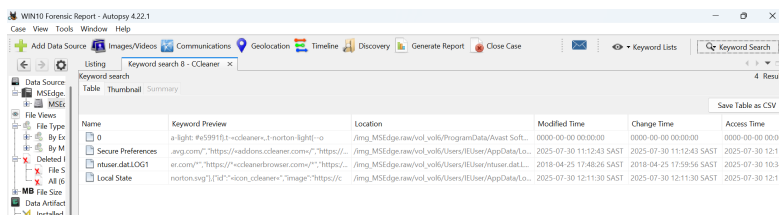


Figure 15: "CCleaner" Key Search Output1

Name	Keyword Preview	Location	Modified Time	Change Time
f0dbba63ec90bc3661a98d1f747295bc44697b897	heatinghutch.magic-erasersearchpy42.sfrbylmeosd...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 11:08:49 SAST	2025-07-30 11:08:49 SAST

Hex	Test	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Content	Annotations	Other Occurrences
-----	------	-------------	---------------	------------	----------------	------------------	---------	-------------	-------------------

Metadata

Name: /img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Local/Google/Chrome/User Data/component_crs_cache/f0dbba63ec90bc3661a98d1f747295bc44697b8974b39d18133ca206c70dec

Type: File System

MIME Type: application/x-chrome-package

Size: 7931809

File Name Allocation: Allocated

Metadata Allocation: Allocated

Modified: 2025-07-30 11:08:49 SAST

Accessed: 2025-07-30 11:08:49 SAST

Created: 2025-07-30 11:08:49 SAST

Changed: 2025-07-30 11:08:49 SAST

MDS: Not calculated

SHA-256: Not calculated

Hash Lookup Results: UNKNOWN

Internal ID: 13808

From The Sleuth Kit istat Tool:

Figure 19: "eraser" Key Search Output1

WIN10 Forensic Report - Autopsy 4.22.1

Case View Tools Window Help

Keyword search 14 - eraser x Keyword search 15 - eraser x Keyword search 16 - eraser x

18 Results

Name	Keyword Preview	Location	Modified Time	Change Time
UseClass.dat-slack	ndrive-appmanagement-vppreviewviewersynchost...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-18 13:06:42 SAST	2018-04-25 17:29:4
modswatch	-lastusederaser	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2018-04-12 11:19:55 SAST	2018-04-25 18:44:1
StructuredQuerySchema.bin	<item document.datonatedsystem.document.datpt...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 12:04:55 SAST	2025-07-30 12:04:5
hthavogtmap@z@z+eraserect+@painter@q...	hthavogtmap@z@z+eraserect+@painter@q...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 10:53:29 SAST	2025-07-30 10:53:2
IAF0698E-0558-476E-983C-3716689AF493.2.ver	m-photo.cameramodel-system.photo.cameraseriahu...	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2025-07-30 11:40:53 SAST	2025-07-30 11:40:5
modswatch	-lastusederaser	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2018-04-12 11:19:55 SAST	2018-04-25 18:44:1
HD08enad-slack	m-photo.cameramodel-system.photo.cameraseriahu...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 12:04:41 SAST	2025-07-30 12:04:4
DD0571F2-8E98-426D-8288-1A9A39C3FDA2.2.w	m-photo.cameramodel-system.photo.cameraseriahu...	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2025-07-30 11:40:53 SAST	2025-07-30 11:40:5
0883C389-8905-4D3D-8BC9-863C603DA134.3.w	<item document.datonatedsystem.document.datpt...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 12:04:56 SAST	2025-07-30 12:04:5
Microsoft.ReactNative.dll	resrecoivouches-riasesen-topouchcancelopt	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 10:53:28 SAST	2025-07-30 10:53:2
modswatch	-lastusederaser	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2018-04-12 11:19:55 SAST	2018-04-25 18:44:1
MicrosoftOffice2016Win32.xml	intelligence_feedingspenseserver.combasic_ads_block...	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2025-07-30 11:12:06 SAST	2025-07-30 11:12:0
Se4361AD-7aa-46b7-917e-3988857b7e4.tmp-4	modswatch	/img_MSEdge.raw/vol_vols/ProgramData/Microsoft...	2018-04-12 11:19:55 SAST	2018-04-25 18:44:1
MicrosoftOffice2016Win64.xml	lyoutitemcommaglnstobkewasnameinktoobafly...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 10:53:30 SAST	2025-07-30 10:53:3
Resources.pri	protector spray+reserpy+ womenrowby boots	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 11:08:49 SAST	2025-07-30 11:08:4
f0dbba63ec90bc3661a98d1f747295bc44697b897	ig@q@ba.nabau12@z@z+eraserect+@painter@q...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 10:53:29 SAST	2025-07-30 10:53:2
Q5Gul.dll	rotation_1.pdfl-vpdf_ink_module.cocstarttestelecto...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 11:05:10 SAST	2025-07-30 11:05:1
Q5054e4-257c-4852-ae3c-202503a3298.tmp	le as a property of <pointevent.qml.addedminionver...	/img_MSEdge.raw/vol_vols/Users/ElUser/AppData/Lo...	2025-07-30 10:53:29 SAST	2025-07-30 10:53:2
Q5Quick.dll				

Figure 20: "eraser" Key Search Output2

WIN10 Forensic Report - Autopsy 4.22.1

Case View Tools Window Help

Keyword search 14 - eraser x Keyword search 15 - eraser x Keyword search 16 - eraser x

18 Results

Time	Size	Flags(Dir)	Flags(Meta)	Known	MDS Hash	SHA-256 Hash	MIME Type	Extension
25 17:48:26 SAST	180024	Allocated	Allocated	unknown			application/octet-stream	dat-slack
12 01:35:10 SAST	68367	Allocated	Allocated	unknown				xml
25 17:49:49 SAST	419771	Allocated	Allocated	unknown			application/octet-stream	bin
30 10:53:29 SAST	5327912	Allocated	Allocated	unknown			application/vnd-msdownload	dll
30 11:40:53 SAST	220464	Allocated	Allocated	unknown			application/octet-stream	db
12 01:35:10 SAST	68367	Allocated	Allocated	unknown	75795f4a04b25ba6c21652903e1ee91	0ee179a0451d41036f8b8996353478f58b533412a6...	application/octet-stream	xml
25 17:58:00 SAST	3194880	Allocated	Allocated	unknown			application/octet-stream	hwd-slack
30 11:40:53 SAST	61912	Allocated	Allocated	unknown	809baeb51385e5910a03c522b8930f	06074f756e59f36e1f9c5111eebf46536ba670234e...	application/octet-stream	db
25 17:51:13 SAST	419808	Allocated	Allocated	unknown			application/octet-stream	db
30 10:53:29 SAST	3080568	Allocated	Allocated	unknown			application/vnd-msdownload	dll
12 01:25:50 SAST	63721	Allocated	Allocated	unknown				xml
30 11:12:06 SAST	502090	Allocated	Allocated	unknown			application/octet-stream	tmp-slack
12 01:35:10 SAST	65734	Allocated	Allocated	unknown				xml
26 20:20:28 SAST	1896240	Allocated	Allocated	unknown			application/octet-stream	pri
30 11:08:49 SAST	7931809	Allocated	Allocated	unknown				
30 10:53:29 SAST	6790032	Allocated	Allocated	unknown			application/vnd-msdownload	dll
30 11:05:10 SAST	1951144	Allocated	Allocated	unknown			application/octet-stream	tmp-slack
30 10:53:29 SAST	4178816	Allocated	Allocated	unknown			application/vnd-msdownload	dll

Figure 21: "eraser" Key Search Output3

Suspicious activity began on **2018-01-25** with anti-forensic behavior, escalated through malware propagation in August 2018, and resurfaced in July 2025 with

coordinated downloads, deletion, and privacy tool usage. The timeline reflects both infection and intentional concealment.

from YTS.mx, a known torrenting platform. Additionally, the user accessed a GitHub-hosted HTML file referencing EternalRocks, indicating interest in or malware research. VPN-related activity was confirmed through searches and downloads from openvpn.net, windscribe.com, and nordvpn.com, suggesting efforts to anonymize browsing, likely to avoid detection while torrenting or accessing restricted content.

Source Name	S	O	Path	URL
2d7104db-30ee-4b4b-b227-cho0648112b.tmp.Zo			/Users/ElUser/AppData/Local/Temp/2d7104db-30ee-4b4b-b227-cho0648112b.tmp.Zo	about:internet
2465d296-d54c-44bd-8c57-31a6a6b6c5.tmp.Zo			/Users/ElUser/AppData/Local/Temp/2465d296-d54c-44bd-8c57-31a6a6b6c5.tmp.Zo	about:internet
319d344-a5d9-4d61-837d-7464e7621717.tmp.Zo			/Users/ElUser/AppData/Local/Temp/319d344-a5d9-4d61-837d-7464e7621717.tmp.Zo	about:internet
b4d49a74-1336-48d4-b6f7-87af2e75ab42.tmp.Zo			/Users/ElUser/AppData/Local/Temp/b4d49a74-1336-48d4-b6f7-87af2e75ab42.tmp.Zo	about:internet
b862118c-fa9a-42d8-a763-486f5044b6c6.tmp.Zo			/Users/ElUser/AppData/Local/Temp/b862118c-fa9a-42d8-a763-486f5044b6c6.tmp.Zo	about:internet
1cca01e7-f7d9-4d4c-bd49-1a0553a4a43a.tmp.Zo			/Users/ElUser/AppData/Local/Temp/1cca01e7-f7d9-4d4c-bd49-1a0553a4a43a.tmp.Zo	about:internet
43317e18-a98a-4111-8191-a5bd786d351.tmp.Zo			/Users/ElUser/AppData/Local/Temp/43317e18-a98a-4111-8191-a5bd786d351.tmp.Zo	about:internet
73516d0e-a916-414e-83d7-8a6d6f155b3d.tmp.Zo			/Users/ElUser/AppData/Local/Temp/73516d0e-a916-414e-83d7-8a6d6f155b3d.tmp.Zo	about:internet
ab4d44ea-d9e7-4215-95ad-8494346b5241.tmp.Zo			/Users/ElUser/AppData/Local/Temp/ab4d44ea-d9e7-4215-95ad-8494346b5241.tmp.Zo	about:internet
Captain America Civil War (2016) [YTS.AG].torrent.Zo			/Users/ElUser/AppData/Roaming/torrent/Captain A...	about:internet
Iron Man (2008).torrent.Zone.Identifier			/Users/ElUser/AppData/Roaming/torrent/Iron Man L...	about:internet
28 Years Later (2025) [720p] [WEBRip] [YTS.MX].torrent			/Users/ElUser/Downloads/28 Years Later (2025) [720p]...	about:internet
70edc2b69f8b5461ba57fbaf55b3d62f8e7c			/Users/ElUser/Videos/70edc2b69f8b5461ba57fbaf55b3d62f8e7c	about:internet
Captain America Civil War (2016) [720p] [BluRay] [YTS.MX].torrent			/Windows/Captain America Civil War (2016) [720p] [B...	about:internet
Iron Man (2008) [720p] [BluRay] [YTS.MX].torrent.Zo			/Windows/Iron Man (2008) [720p] [BluRay] [YTS.MX]...	about:internet
MicrosoftEdgeCookiesBackup.dat.Zone.Identifier			/Users/ElUser/MicrosoftEdge/backups/backups/Micros...	about:internet
MicrosoftEdgeSettingsBackup.btt.Zone.Identifier			/Users/ElUser/MicrosoftEdge/backups/backups/Micros...	about:internet
Backup.dat.Zone.Identifier			/Users/ElUser/MicrosoftEdge/backups/backups/Micros...	about:internet
ubuntu-25.04-desktop-amd64.iso.torrent.Zone.Ide		4	/Users/ElUser/AppData/Roaming/torrent/ubuntu-25...	https://releases.ubuntu.com/25.04/ubuntu-25.04-c...
ubuntu-25.04-desktop-amd64.iso.torrent.Zone.Ide		4	/Users/ElUser/Downloads/ubuntu-25.04-desktop-am...	https://releases.ubuntu.com/25.04/ubuntu-25.04-c...
ubuntu-22.04.5-live-server-amd64.iso.torrent.Zone		4	/Users/ElUser/Downloads/ubuntu-22.04.5-live-serve...	https://releases.ubuntu.com/22.04/ubuntu-22.04.5...
transmission-4.0.6-a64 (1).msi.Zone.Identifier		3	/Users/ElUser/Downloads/transmission-4.0.6-a64 (1)...	https://releases-assets.githubusercontent.com/gith...

Figure 24: Web Downloads Output

4.3. Browser Cookies (Saved table as .csv, then used AI to analyze it)

Although no direct IP addresses were recovered, several artifacts imply outbound network activity. These include the download of a VPN client installer (`openvpn-connect-3.7.3.4351_signed.msi`), Bing searches such as “how to torrent without getting caught,” and DNS error pages (`dnserror.html`) that suggest failed or redirected network attempts. These behaviors collectively point to attempts at anonymization and possible efforts to bypass network restrictions.

Cookies extracted from `WebCacheV01.dat` show consistent interaction with domains such as `bing.com`, `msn.com`, `live.com`, `youtube.com`, `linkedin.com`, and `google.com`, reflecting general browsing, search activity, and Microsoft account usage. More notably, cookies from `nordvpn.com` and `windscribe.com` confirm active VPN usage through session identifiers and referral metadata. These VPN services are commonly used to mask identity during torrenting or forum access. No cookies were found from known hacking forums such as `nulled.to`, `exploit.in`, or `breachforums.is`. However, indirect indicators, including GitHub malware references, torrenting behavior, and proxy artifacts such as `localhost:57585/proxy/0/`, suggest intent to anonymize and possibly engage

in suspicious activity.

Win10 Forensic Report - Autopsy 4.22.1

Case View Tools Window Help

Web Cookies

600 Results

Source Name	S	C	O	URL	Name	Value	Program Name	Domain	Data Source
WebCacheV01.dat	3			c.dailym...	MR	0	Microsoft Edge Analyzer	dailym...	MSEdge.raw
WebCacheV01.dat	3			c.dailym...	ANONCHK	0	Microsoft Edge Analyzer	dailym...	MSEdge.raw
WebCacheV01.dat	3			www.bing.com	MSA.Auth.Nonce.C...	N	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			www.bing.com	MSA.Auth.Nonce.C...	N	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			nordvpn.com	experiment_local	g1R2D.1	Microsoft Edge Analyzer	nordvpn.com	MSEdge.raw
WebCacheV01.dat	3			nordvpn.com	nc	17539006...	Microsoft Edge Analyzer	nordvpn.com	MSEdge.raw
WebCacheV01.dat	3			nordvpn.com	nci	30	Microsoft Edge Analyzer	nordvpn.com	MSEdge.raw
WebCacheV01.dat	3			nordvpn.com	bs	95	Microsoft Edge Analyzer	nordvpn.com	MSEdge.raw
WebCacheV01.dat	3			nordvpn.com	...cf_bm	tuqV0xW...	Microsoft Edge Analyzer	nordvpn.com	MSEdge.raw
WebCacheV01.dat	3			www.bing.com	MSA.Auth.Nonce.C...	N	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			windscribe.com	pk_4f3.2e1e	N5SR6278...	Microsoft Edge Analyzer	windscribe.com	MSEdge.raw
WebCacheV01.dat	3			windscribe.com	pk_4f3.2e1e	25a02a06...	Microsoft Edge Analyzer	windscribe.com	MSEdge.raw
WebCacheV01.dat	3			windscribe.com	pk_ses.3.2e1e	1	Microsoft Edge Analyzer	windscribe.com	MSEdge.raw
WebCacheV01.dat	3			youtube.com	_Secure-ROLLOUT...	CL2hgbbu...	Microsoft Edge Analyzer	youtube.com	MSEdge.raw
WebCacheV01.dat	3			youtube.com	VISITOR_INFO_LIVE	[B8xZh8Q]	Microsoft Edge Analyzer	youtube.com	MSEdge.raw
WebCacheV01.dat	3			youtube.com	VISITOR_PRIVACY...	CglaQR6G...	Microsoft Edge Analyzer	youtube.com	MSEdge.raw
WebCacheV01.dat	3			google.com	NID	525-HyR...	Microsoft Edge Analyzer	google.com	MSEdge.raw
WebCacheV01.dat	3			www.bing.com	MSA.Auth.Nonce.C...	N	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			bing.com	SRCHPGUSR	HV-17538...	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			bing.com	_clk	1p8mmus...	Microsoft Edge Analyzer	bing.com	MSEdge.raw
WebCacheV01.dat	3			support.google.com	_ga_H3OR8PNDQFN	652.1s175...	Microsoft Edge Analyzer	google.com	MSEdge.raw

Figure 25: Browser Cookies Output

Task 5: Email or Messaging Artifact Recovery

1. Skype Filters

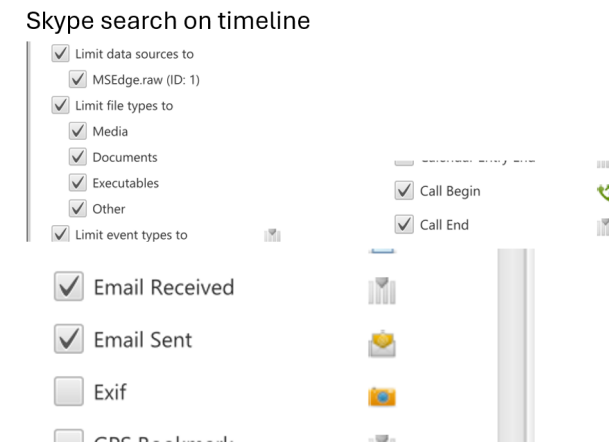


Figure 26: Skype filters chosen during artifact recovery

2. Skype Timeline and Usage

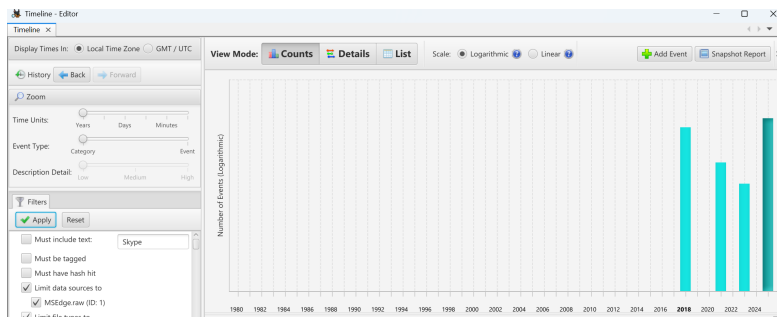
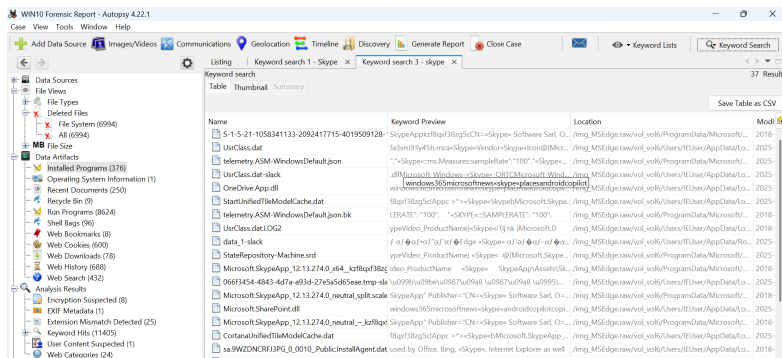
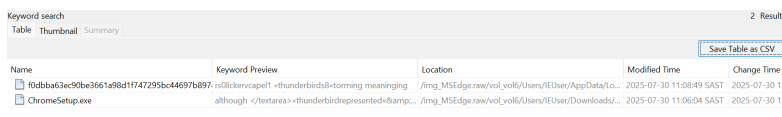
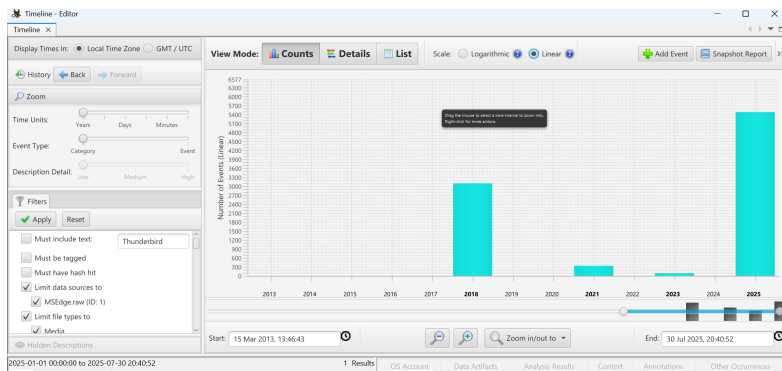


Figure 27: Skype use timeline output



3. Thunderbird



4. Outlook

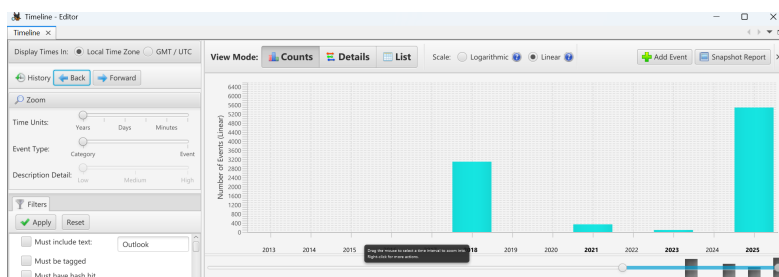


Figure 31: Outlook use timeline output

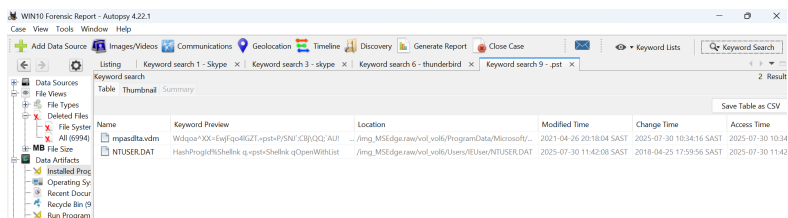


Figure 32: Evidence of Outlook usage via ".pst" key search

5. Suspicious Emails

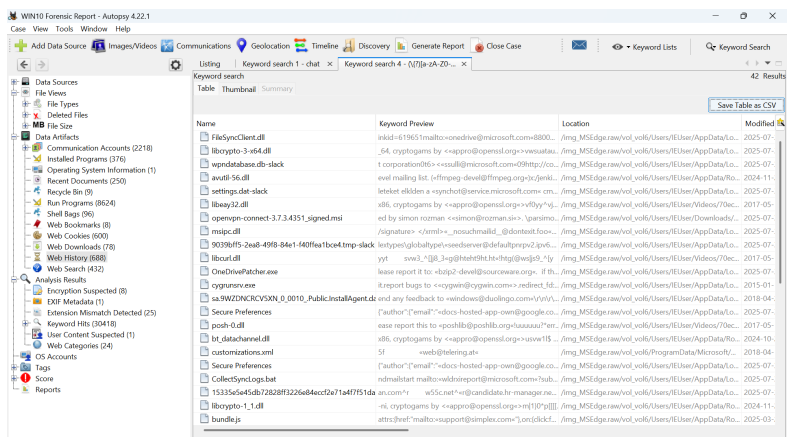


Figure 33: Suspicious email evidence via regular expression

The screenshot shows the Autopsy 4.22.1 interface with a keyword search for "email". The search results are displayed in a table with columns: Name, Keyword Preview, Location, and Modified. The table lists various files and folders, including "misp_protection_sdk.dll", "UserClass.dat", "ibcrypt-3-v64.dll", "OneDrive.App.dll", "UserClass.dat.LOG1", "UserClass.dat.LOG2", "UserClass.dat.LOG2-slack", "mpadesc.dll.mui", "Web Categories Artifact", "login.live.comName: Web <email>", "databases.def", "helper.exe", "ibcrypt-1.1.dll", "Statefulpostitory-Deployment.asd", "S-1-5-21-1058341133-2092417715-4019509128", "HubApps", "WebCacheV01.dat-slack", and "CrashpadMetrics.pma-slack". The "Keyword Preview" column shows snippets of text from the files, often containing email addresses or related information. The "Location" column shows the file's path within the forensic image.

Name	Keyword Preview	Location	Modified
misp_protection_sdk.dll	not found is not a valid <email> addressParameter na...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
UserClass.dat	ms-unistore-<email>URLms-unistore-<email>ok SAApp...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
ibcrypt-3-v64.dll	hostname(s) = Expected <email> address = %sExpected...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
OneDrive.App.dll	SignIn EnabledHRD Minimal <Email> Validation Enabl...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
UserClass.dat.LOG1	antivirusweb protection-<email> protectionWindows...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2018-04-
UserClass.dat.LOG2	antivirusweb protection-<email> protectionWindows...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2018-04-
UserClass.dat.LOG2-slack	antivirusweb protection-<email> protectionWindows...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2018-04-
mpadesc.dll.mui	WormbackdoorTrojan-Email- FlooderDataMonitoring	/img_MSIEdge.raw/vol_vo/0/ProgramData/Microsoft/...	2025-07-
Web Categories Artifact	login.live.comName: Web <email>	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
databases.def	ato.com["banking"] "<email>-casco.com["banking"]	/img_MSIEdge.raw/vol_vo/0/ProgramData/Avast Soft...	2025-07-
helper.exe	Information Updating <email> addressDjenkins wor...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
ibcrypt-1.1.dll	ClientSSL ServerNAME <email>Object SignatCCP m...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2024-11-
Statefulpostitory-Deployment.asd	col Name: "ms-unistore-<email>" [img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/Roaming/Usentent Web/ibcrypt-1.1.dll	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
S-1-5-21-1058341133-2092417715-4019509128	CorporationRESA&femail and accountsfemail and...	/img_MSIEdge.raw/vol_vo/0/ProgramData/Microsoft/...	2018-04-
HubApps	"text": "View source <email>" }	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
WebCacheV01.dat-slack	persistent handler for mapi <email>ok cjsd767ehd-cc...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
CrashpadMetrics.pma-slack	"Pinakatumdigma para sa mapi <email>"; "Pinakatum...	/img_MSIEdge.raw/vol_vo/0/Users/ElUser/AppData/lo...	2025-07-
msnashar &femail	WormbackdoorTrojan-Email- FlooderDataMonitoring	/img_MSIEdge.raw/vol_vo/0/ProgramData/Microsoft/...	2018-04-

Figure 34: Suspicious email evidence via "email" key search

Task 6: Suspicious Software or Malware Evidence

6.1. Suspicious Executables or Tools

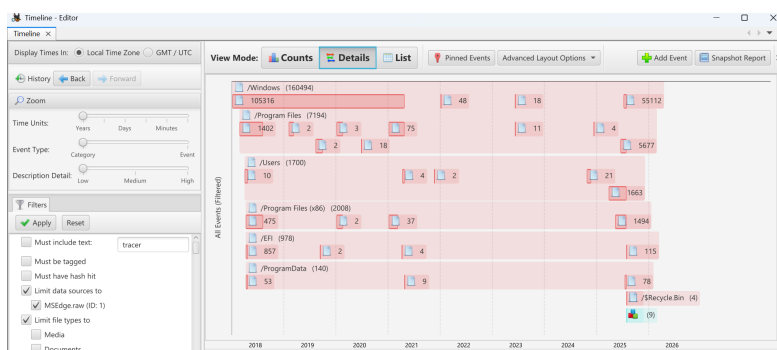


Figure 35: Suspicious executables identified using "trace" key search

6.2. Hash Verification and Threat Analysis



Figure 36: SHA-256 Hash: 28c03d41ec28365f9cb58c1281b29d7e6ca64decf77e2ba52807f4a4914e038a

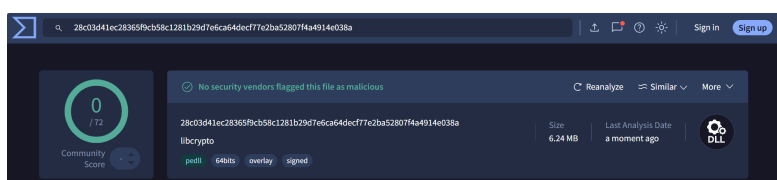


Figure 37: Threat analysis result: No security vendors flagged this file

Conclusion and Recommendations

The analysis revealed the presence of suspicious executables, VPN activity, and browser traces linked to torrenting and malware. Registry entries and startup

configurations indicated attempts to maintain persistence and conceal digital footprints. Based on these findings, the case should be escalated for disciplinary review. The compromised system must be reimaged, and stricter network and software policies should be enforced. Additionally, cybersecurity awareness training is recommended to prevent future incidents. These actions will strengthen digital oversight and uphold institutional integrity.